Viva and demonstration guide

Stroke Risk Factor Analysis Using Patient Health Records

A five-minute demonstration

0:00–0:40 · Problem and source

State the five factors, show stroke.csv and the Kaggle source. Explain the 5,110 rows, 12 columns and binary outcome.

0:40–1:30 · Cleaning

Show validation functions, missing BMI and Unknown smoking. Explain the adult primary population and why raw records are preserved.

1:30–2:30 · Visual results

Show age, condition and smoking charts. Read the counts and confidence intervals, not only the bar heights.

2:30–3:40 · Statistics

Show primary_test_family.csv. Explain Welch, chi-square, Holm correction and the BMI null result.

3:40–4:30 · Sensitivity

Explain why BMI changes with children included and why smoking changes in complete adult records.

4:30–5:00 · Limitations

Mention provenance, missingness, confounding and no causal or future-risk claim.

Command sequence

```
python main.py
python verify_project.py
python build_documents.py
```

Know these numbers

Raw: 5,110 records; 249 stroke labels (4.87%). Adults: 4,254; 247 stroke labels. Adult BMI missing: 181. Adult smoking unknown: 862. Complete adults: 3,256.

Questions about data and methods

Why did you select this dataset?

It includes all five requested factors, a binary stroke label, manageable CSV size and enough data-quality issues to demonstrate an end-to-end analysis.

Why exclude children from the main tests?

Adult BMI and smoking comparisons are easier to interpret without mixing pediatric records. Children remain in the raw data and the all-age sensitivity analysis.

Why not replace Unknown with never smoked?

Unknown does not establish an absence of smoking. It remains separate in descriptions and is excluded from the known-status test.

How did you handle missing BMI?

Primary tests use observed BMI only. A separate optional export uses the adult median and flags filled values, but the tests never use it.

Why Welch instead of a pooled t-test?

The groups have very unequal sizes and may have different variances. Welch estimates the standard error and degrees of freedom without assuming equal variances.

Why chi-square for smoking and hypertension?

Both the factor and outcome are categorical. We compare observed table counts with those expected under independence; the minimum expected counts exceed five.

Why correct multiple tests?

Five tests create more chances of false positives. Holm adjusts their p-values as one primary family while controlling family-wise error under its assumptions.

What is the confidence interval?

The procedure has 95% coverage over repeated comparable samples under its assumptions. It is not a 95% probability statement about a fixed parameter after observing this interval.

Questions about results and limitations

What are your strongest observations?

Age and average glucose differ substantially between outcome groups; hypertension also shows an association. These are unadjusted associations, not independent causal effects.

What did you find for BMI?

Adult mean difference is about 0.04 kg/m², with CI −0.86 to 0.94 and adjusted p about 0.933. We did not detect an adult mean difference; this does not prove no relationship.

Does your smoking chart say quitting is harmful?

No. Former smokers are older and smoking timing and dose are unavailable. The small association changes with the analysis population. The chart cannot support a causal interpretation.

Why is all-age BMI significant but adult BMI is not?

The result is sensitive to population composition. Mixing children with adults changes the comparison. We cannot treat the all-age result as an independent BMI effect.

Would 95% classification accuracy prove a good model here?

No. A classifier predicting no stroke for everyone would already score about 95.13% in the raw dataset. We do not train or claim performance for a predictive model.

What is Hedges g?

The difference in group means divided by pooled standard deviation with a small-sample correction. It complements p-values; the Welch CI remains the primary interval for the mean difference.

What is the largest limitation?

Source provenance and sampling are incompletely verified. Missingness differs by outcome, and covariates are not jointly adjusted. The findings cannot be generalized as population risk estimates.

What would you improve next?

Obtain well-documented data with dates, assess missing-data assumptions and fit an adjusted model. If predicting future outcomes, add appropriate validation and calibration.